
mmBERT: A Modern Multilingual Encoder with Annealed Language Learning

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Abstract

Encoder-only language models are frequently used for a variety of language tasks, including classification and retrieval. However, there has been a lack of recent research efforts for encoder models, especially with respect to multilingual models. We introduce MMBERT, an encoder-only language model pretrained on 3T tokens of multilingual text in over 1800 languages. To build MMBERT we introduce several novel elements to massively multilingual encoder training, including phased data curation and scheduled language inclusion. We add over 1700 low-resource languages to the data mix only during the decay phase, showing that it boosts performance dramatically and maximizes the gains from the relatively small amount of training data without excessive repetition. The model uses recent advances in architecture and training schemes to be faster and more multilingual than other models and we release weights, data, and code. We show that MMBERT significantly outperforms the previous generation, on various tasks, for both high and low-resource languages.

1. Introduction

Encoder-only language models (LMs) were developed during the early years of scaling language model pre-training, including models such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). These models were followed by multilingual variants such as mBERT (Devlin et al., 2019) and XLM-RoBERTa (also known as XLM-R) (Conneau et al., 2019). However, these models cannot generate text, and thus have fallen out of popularity in favor of larger

decoder-only language models (Brown et al., 2020; Achiam et al., 2023; Team et al., 2023; Dubey et al., 2024).

Despite this, encoder-only models still are frequently used for natural language understanding (NLU) tasks, including classification, clustering, and retrieval. For many years, these older models were still the best encoder-only models available. However, there has been a recent revival of encoder-only model pretraining (Nussbaum et al., 2024; Portes et al., 2023; Warner et al., 2024; Boizard et al., 2025), bringing modern pre-training techniques for decoder-only language models to encoder-only models. There has also been recent work showing that encoder models are significantly better for classification/retrieval than decoder models for a given size, even those an order-of-magnitude larger (Weller et al., 2025; Gisserot-Boukhlef et al., 2025).

In this encoder-only model revival there has been a conspicuous lack of large-scale multilinguality. Although it is over six years old, XLM-R is still often SoTA – especially surprising when you consider the fast-paced nature of the LM field. Thus, we aim to provide an improved model.

We do this by pre-training our new model suite, MMBERT¹, on 3T tokens of multilingual text. We add novel components to the pre-training recipe: updated data and modeling choices, an inverse mask schedule (high $\Rightarrow$ low), an annealed language schedule (more biased $\Rightarrow$ more uniform), and phased training with an increased number of languages (60 $\rightarrow$ 110 $\rightarrow$ 1833), allowing for maximal impact of the smaller amount of data.

MMBERT improves over XLM-R across the board, and even beats models like OpenAI’s o3 (OpenAI, 2025) and Google’s Gemini 2.5 Pro (Comanici et al., 2025) on low-resource languages. We show that including the low-resource languages in the decay phase enables rapid learning, boosting performance on these languages roughly 2x despite only using 100B tokens. Overall, MMBERT is the first model to show significant improvements over XLM-R for massively multilingual data while also introducing new techniques for multilingual pre-training.

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¹Models, data, and code are available at [HuggingFace](#)

Table 1. Configuration for MMBERT models (small is 140M params, base is 307M). MMBERT small initialized from Base and required LR/WD reduction to 4e-4/4e-5 after 1.2T tokens.

Model	Layers	Hidden Size	Intermediate Size	Heads	LR	WD	Warmup (B)	BS Warmup (B)
MMBERT small	22	384	1152	6	8e-4	8e-5	4	100
MMBERT base	22	768	1152	12	8e-4	8e-5	3	60

Shared Parameters:

Vocabulary Size: 256,000 | Training Batch Size: 4.7M tokens | Max Sequence Length: 1024 → 8192

RoPE Base: 10k → 160k | Sliding Window: 128 | Global Attention Every N Layers: 3

Masking Rate Schedule: 30% → 15% → 5% | Tokenizer: Gemma 2 | Architecture: ModernBERT

2. Related Work

Encoder-only Models Encoder-only models were the predominant language model in the early LM days, with ELMo (Peters et al., 2018), BERT (Devlin et al., 2019), and RoBERTa (Liu et al., 2019) being early examples of scaling language models up to the trillion token data range. Encoder-only models are still the predominant models used for classification and retrieval tasks when inference speed is important. However, decoder-only models have been scaled to the trillion parameter scale whereas encoder models typically remain less than 1 billion parameters.

The revival of encoder-only LM development was spurred by works such as MosaicBERT showing a BERT-equivalent model could be trained in under 24 hours (Portes et al., 2023; Nussbaum et al., 2024). More recently, ModernBERT (Warner et al., 2024) further scaled these recipes and showed that you could greatly improve performance. Since then there have been several more: EuroBERT focusing on 15 languages (Boizard et al., 2025), NeoBERT (Le Breton et al., 2025) and Ettin (Weller et al., 2025) on English. However, none of these have scaled to more than 15 languages.

In the massively multilingual setting, there are very few available models: mBERT with 104 languages (Devlin et al., 2019), XLM-R with 100 languages (Conneau et al., 2019), and mGTE with 74 languages (Zhang et al., 2024). Works such as multilingual DistilBERT (Sanh et al., 2019) and multilingual MiniLM (Wang et al., 2020) have distilled from mBERT and XLM-R respectively to derive smaller variants. The more recent mGTE showed slightly improved performance over XLM-R while allowing for longer contexts, while mBERT is generally not used due to the improvements of XLM-R. Despite XLM-R’s release date, it has aged very well: its design was well ahead of its era through its use of 6T training tokens, more than any other encoder-only model has ever been trained on, even to this day (including our MMBERT). However, data quality has significantly increased since 2019, allowing us to achieve higher scores with only half the tokens (Penedo et al., 2024).

Multilingual LMs and Datasets Multilingual models and datasets have a long history of development for machine translation systems, ranging from bilingual models and datasets to massively multilingual systems (NLLB Team et al., 2022; Arivazhagan et al., 2019, among others). Recently, decoder-only LMs (Gemma et al., 2024; Dubey et al., 2024) have embraced multilinguality in order to increase accessibility (Comanici et al., 2025; Achiam et al., 2023). However, many of the details of these models are not available, including pre-training data. This has created an access gap between open massively multilingual systems and closed systems. From models that do release information, we know that improvements are often due to improved data quality or techniques like parallel data (Martins et al., 2025; Li et al., 2024b; Dubey et al., 2024). For massively multilingual datasets, previous work often derived from Common-Crawl (Wenzek et al., 2019). Our work also uses datasets derived from web-scale corpora, but with recent advances in filtering and language identification (Penedo et al., 2025; Kargaran et al., 2023) and other datasets from specialized sources.

3. Training Details

3.1. Architecture

We use an identical architecture to ModernBERT (Warner et al., 2024), but employ the Gemma 2 tokenizer² (Gemma Team et al., 2024) to handle multilingual input. We use 22 layers and a 1152 intermediate dim for both base and small versions (same as ModernBERT-base), but use a hidden dimension of 768 for base and 384 for small. For the rest of the training configurations that are shared, see Table 10 in the Appendix.

Our base version has the same number of non-embedding parameters as ModernBERT-base (110M) but a total of 307M

²During training of MMBERT the Gemma 3 tokenizer was released. We would encourage future work to pre-train using this tokenizer after modifying the pre-tokenizer to include prefix spaces, which we did not use. This likely would help for NER/POS.

Table 2. Training data mixture across the various training stages (pre-training, mid-training, decay). Later stages use higher quality data, including the recent Dolmino (OLMo et al., 2025) and FineWeb2-HQ (Messmer et al., 2025) datasets. Dashes indicate that no data from that source was used. We trained for 2.3T tokens for pre-training, 600B for mid-training (e.g. context-extension to 8192 sequence length), and 100B for the decay phase. We sample from the dataset and repeat (or under-sample) as needed to hit the token counts used for training. Note that the decay phase included three different mixtures and the resulting weights were merged together, this includes only one version (Decay-Cont). See Appendix D for more details. If not specified, the source is Dolma v1.7.

Category	Dataset	Pre-training		Mid-training		Decay Phase	
		Tokens (B)	%	Tokens (B)	%	Tokens (B)	%
Code	Code (ProLong)	–	–	–	–	2.8	2.7
Code	StarCoder	100.6	5.1	17.2	2.9	0.5	0.5
Crawl	DCLM	600.0	30.2	10.0	1.7	–	–
Crawl	DCLM (Dolmino)	–	–	40.0	6.7	2.0	2.0
Crawl	FineWeb2	1196.6	60.2	506.7	84.3	78.5	76.0
Instruction	Tulu Flan	15.3	0.8	3.1	0.5	1.0	1.0
Math	Dolmino Math	11.2	0.6	4.3	0.7	0.5	0.5
Reference	Books	4.3	0.2	3.9	0.7	2.2	2.1
Reference	Textbooks (ProLong)	–	–	–	–	3.1	3.0
Reference	Wikipedia (MegaWika)	4.7	0.2	1.2	0.2	9.5	9.2
Scientific	Arxiv	27.8	1.4	5.4	0.9	3.3	3.2
Scientific	PeS2o	8.4	0.4	3.2	0.5	–	–
Social	StackExchange	18.6	0.9	3.0	0.5	–	–
Social	StackExchange (Dolmino)	1.4	0.1	2.8	0.5	–	–
Total		1989.0	100.0	600.8	100.0	103.3	100.0

parameters due to the larger vocabulary. MMBERT small has 140M total parameters, with 42M non-embedding parameters. As we show in (§4.5) these architectures are also significantly faster than any previous multilingual encoder model while being the same standard sizes.

3.2. Training Data

We follow the Ettin recipe (Weller et al., 2025) as the only open-data ModernBERT equivalent. Crucially though, we change the source of web crawl to account for more multilingual data. Previous work such as XLM-R and mT5 (Xue et al., 2020) used a very low percentage of English content (5.7% for mT5). However, the highest quality data (filtered DCLM (Li et al., 2024a; OLMo et al., 2025)) only exists in English. Thus we choose to use a significantly higher percentage of English compared to previous work (from 10% to 34% depending on the stage, see Table 11). Despite this relatively high amount of English, we see both strong English performance as well as strong multilingual performance compared to massively multilingual encoders.

Nonetheless, a significant portion of our training data was non-English: for this we used existing data from FineWeb2 (Penedo et al., 2025) and an existing filtered version of 20 languages from FineWeb2 called FineWeb2-HQ (Messmer et al., 2025). We also use and filter MegaWika v2 (Barham

et al., 2023; 2025) for Multilingual Wikipedia, which covers 60 languages (those from our first stage).

We include several other curated corpora in English. From Dolma (Soldaini et al., 2024) we use StarCoder, StackExchange, Arxiv, and PeS2o. From Dolmino (OLMo et al., 2025) we use math, filtered StackExchange (which is mainly code), Tulu Flan instruction data, and books (OLMo et al., 2025). From ProLong (Gao et al., 2024) we use their code repositories and their textbooks. All our data is publicly available on HuggingFace.³ Thus, our data mix is higher quality than previous work (through the use of filtered DCLM and FineWeb2), more diverse in content (code, instructions, web data, papers), and includes a greater variety of languages and scripts.

Annealed Language Learning Previous work typically used a fixed set of languages for training multilingual models. However, in the highly multilingual setting, this forces a tradeoff. The data mixture must either upsample (repeat) low-resource data, or accept that low-resource languages will be diluted through training. We propose including the lowest resource languages during only the final portion of training. We motivate this intuition with a synthetically

³We did not create any new datasets for training, however, we did gather them and create the final data mix. We thank the open-source efforts cited above for making this possible.

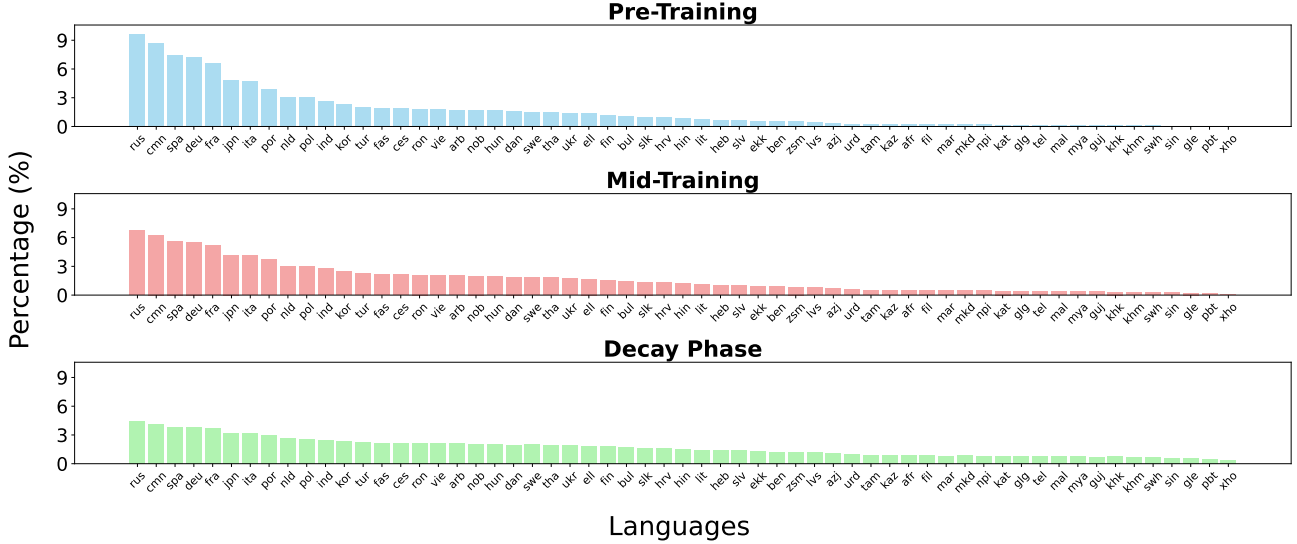


Figure 1. Our inverse temperature sampling ratio throughout training from FineWeb2 data, from τ of 0.7 to 0.5 to 0.3. Note that other sources are excluded from this chart and are not temperature sampled. For full language percent details, see Appendix D. Our training starts out more high-resource biased but becomes increasingly uniform. We also include another 50 languages in mid-training and 1723 more in decay not visualized. The percent of English data (not included in FineWeb2) are 34.5%, 12.1%, and 10.2% respectively.

Table 3. Masked Accuracy results for a constrained annealing experiment. The annealed setting lags until the final stage of training, where the Turkish data is first introduced. The natural setting distributes Turkish data throughout the trajectory.

Duration	5B	10B	15B	20B	25B	30B
annealed	51.56	54.77	56.28	57.24	57.88	69.32
natural	58.91	62.78	64.61	65.63	66.28	68.37

constrained experiment. We construct a model using English, German, and Turkish data. We restrict Turkish to only 100M unique tokens and train two models: one with a *natural* distribution⁴ and one with the *annealed* distribution, where Turkish data is introduced only at the end of training. The training duration is the same, 30B tokens, and we decay the learning rate for the last 5B tokens. In the annealed setting, Turkish data is added only during the 5B decay portion. Results are in Table 3. We use masked accuracy on equal held out data as a simple small-scale metric.

These results suggest that including rare data during the final phase is better than distributing it through training. This may be related to results in decoder model training that show changing the data mixture during learning rate decay can increase performance for monolingual, larger, decoder models (Blakeney et al., 2024). Since there is less data to filter, lower-resource languages are also relatively lower quality than highly filtered data for high-resource languages. We want to learn from all languages while keeping the

⁴Equal parts English and German and 100M tokens of Turkish text. This is roughly the ratio of Turkish to English in our dataset.

overall data quality high. In addition to introducing lower resource languages at the end of training, we also resample the language ratio between stages (Table 2). This takes the distribution from more high-resource biased to more uniform. In this setup, the model can learn new languages quickly as it has a strong base in existing languages (e.g. starting with Icelandic and then quickly learning Faroese).

We start with a set of 60 languages (as well as code) that cover a broad range of language families and scripts. We then increase this to 110 languages, covering more “mid-resource” languages (greater than 200 million tokens of data). Finally, we include all languages/scripts included in FineWeb2 (1833 languages, 1895 language/script variants). Our temperature for sampling languages goes from 0.7 to 0.5 to 0.3 (see Figure 1). Including all the languages at the very end allows us to take advantage of the decay phase learning to rapidly increase performance (Section 4.4).

3.3. Training Recipe

We use the same three phase approach as ModernBERT and Ettin but employ an inverse masking rate learning schedule. Rather than using a lower masking ratio at the end of training as shown by Boizard et al. (2025); Weller et al. (2025), we progressively lower the mask rate during training. This directionally follows Ankner et al. (2024) but we change the mask ratio as a step function alongside data phases.⁵

⁵Due to the scale of pre-training, we were not able to ablate this except in the decay phase. However, we found that the smaller the masking rate the better for the decay phase of training.

Table 4. GLUE (English) benchmark results. We see that MMBERT outperforms all other models for their size, with the base version coming close to even ModernBERT’s performance.

	Model Name	Single Sentence		Paraphrase and Similarity			Natural Language Inference			Avg
		CoLA	SST-2	MRPC	STS-B	QQP	MNLI	QNLI	RTE	
Small	mDistilBERT	34.7	89.4	90.5	87.5	86.6	79.0	87.5	73.3	77.5
	Multilingual MiniLM	25.4	91.6	91.3	88.6	87.7	82.1	89.7	75.8	78.3
	MMBERT Small	61.8	93.1	91.6	90.3	88.6	85.8	91.9	81.9	84.7
Base	EuroBERT 210m	36.8	90.6	92.4	89.8	88.3	85.3	91.3	78.3	81.2
	XLM-R Base	54.2	93.1	92.2	89.4	88.5	85.0	90.6	78.7	83.3
	mGTE Base	54.7	93.3	92.7	89.9	89.0	85.3	91.1	82.3	84.0
	MMBERT Base	61.9	94.0	91.9	91.0	89.4	87.7	93.3	85.6	86.3
	ModernBERT Base	65.3	95.3	90.9	91.5	89.4	88.8	93.7	87.7	87.4

Table 5. XTREME benchmark results. Note that we exclude EuroBERT as it excludes many of the languages tested in this benchmark. For a fair comparison with EuroBERT see Section 4.2 where we compare on EuroBERT-included languages specifically.

	Model Name	Cross-lingual Understanding			Question Answering			Structured Prediction		Avg
		XNLI	PAWS-X	XCOPA	XQuAD	MLQA	TyDiQA	WikiANN	UDPOS	
Small	mDistilBERT	60.8	80.2	52.7	49.4	43.5	44.2	54.5	67.1	56.5
	Multilingual MiniLM	71.2	84.6	59.2	68.2	56.8	63.0	59.2	74.2	67.1
	MMBERT small	73.6	86.7	61.8	73.0	62.5	66.7	54.3	70.6	68.6
Base	XLM-R base	74.6	85.9	61.2	73.4	62.1	70.5	61.4	74.3	70.4
	mGTE Base	73.9	86.4	63.6	75.7	64.3	69.9	60.7	74.3	71.1
	MMBERT Base	77.1	87.7	67.5	77.6	66.0	74.5	58.2	74.0	72.8

For MMBERT small, we start the training process by initializing the weights from the completed base version by using strided weight sampling (Sanh et al., 2019).

Base Pre-training This stage encompasses the warmup and stable phase of the trapezoidal learning rate, training for 2.3T tokens. We use both learning rate and batch size warmup. The data in this stage does not include the filtered FineWeb2 data or the higher quality DCLM. We use just 60 languages (plus code languages) in this stage of training and a 30% masking rate⁶ to start.

Context Extension / Mid-Training Here we increase the quality of the data by switching to the filtered higher-quality versions of the datasets (Table 2). We also change the RoPE values (Su et al., 2024) to handle up to 8192 tokens (i.e. theta of 160k) for global and local layers. We further increase the number of languages to 110 languages (plus code). We train for 600B tokens and continue the stable phase, lowering the mask rate to 15%.

⁶MMBERT small lowered the mask rate to 20% / LR to 4e-4 after 1.2T tokens when it had stopped learning.

Decay Phase Finally, we use an inverse square root learning rate schedule to decay for 100B tokens to 0.02 of the peak LR. Unlike ModernBERT and Ettin, we choose to decay with a 5% mask rate and use three different datasets to produce three different variants: English-focused (Decay-Eng), 110 languages (same as the mid-training phase, Decay-Cont), and a 1833 language variant (all FineWeb2 languages, Decay-All). For specific details on each, see Appendix D.

Model Merging We use model merging to combine the best qualities from each mixture. For the base version, we select the best checkpoint from each mixture and use TIES-merging (Yadav et al., 2023) to mitigate parameter interference. Merging across mixtures was ineffective for small, likely due to less parameter agreement in the smaller weight space. Thus, we merged an exponential weighting of the Decay-All checkpoints, as that mixture performed best.

4. Results

4.1. Benchmark Scores

Datasets We benchmark MMBERT on existing encoder benchmarks for NLU and retrieval: GLUE (Wang et al.,

Table 6. Retrieval scores on the CoIR Benchmark. MMBERT models outperform all others except EuroBERT, which used the higher quality but not publicly accessible Stack v2 training data.

Task (→)		Text-to-Code			Code-to-Text	Code-to-Code			Hybrid Code			Avg
Model (param.) ↓		Apps	CosQA	Synthetic Text2sql	Code SearchNet	SN-CCR	CodeTrans -Contest	-DL	StackOver Flow QA	CodeFeedBack -ST	-MT	
Small	mDistilBERT	2.7	8.4	35.2	69.4	31.0	27.5	21.0	45.8	37.4	22.8	30.1
	Multilingual MiniLM	2.2	7.6	22.7	59.8	21.8	24.9	22.4	40.4	35.6	22.6	26.0
	MMBERT small	4.3	19.5	41.5	64.1	38.4	56.3	32.9	60.3	52.4	40.7	41.0
Base	XLM-R base	3.1	13.2	30.5	75.6	30.7	34.4	22.1	51.2	47.3	28.1	33.6
	mGTE base	3.8	13.2	33.8	74.8	39.1	45.2	30.6	58.0	50.3	39.9	38.9
	MMBERT base	6.1	24.6	48.1	63.8	41.7	54.9	33.0	62.1	55.5	32.4	42.2
	EuroBERT 210m	5.6	20.3	43.0	77.0	43.8	62.2	35.0	64.4	57.2	44.6	45.3

2018), XTREME (Hu et al., 2020), and MTEB (Enevoldsen et al., 2025). We differ from early encoder work by dropping the retrieval section of XTREME as there exist improved retrieval benchmarks from MTEB. We also include the code retrieval benchmark CoIR (Li et al., 2024c), although code is not the primary focus of MMBERT.

Baselines We use a variety of baselines: the older but still ubiquitous XLM-R (Conneau et al., 2019), mGTE (Zhang et al., 2024), and EuroBERT-210m (Boizard et al., 2025) for the base size and mDistilBERT (Sanh et al., 2019) and Multilingual MiniLM (Wang et al., 2020) for the small size.⁷ For English, we also show results with ModernBERT (Warner et al., 2024) as an upper bound. We perform a sweep over hyperparameters for each model (Appendix B).

NLU Results We start with English GLUE in Table 4. MMBERT small performs significantly better than other small variants (84.7 average vs MiniLM’s 78.3) and even outperforms all other previous base-sized models, including XLM-R. MMBERT base outperforms all other multilingual models, while even approaching ModernBERT’s English performance (86.3 MMBERT vs 87.4 ModernBERT), despite using a majority of non-English data.

For multilingual performance in XTREME (Table 5), MMBERT base also outperforms all other models on average (72.8 average vs XLM-R’s 70.4), except for structured prediction where it lags behind on POS and ties on NER. This is likely due to the same problem that ModernBERT has, i.e. the lack of a consistent prefix whitespace token during pre-training. We would recommend future work resolve this issue. However, MMBERT base shows significant improvements in classification (e.g. 77.1 XNLI accuracy vs XLM-R’s 74.6) and in question-answering (74.5 F1 on TyDiQA vs 70.5 XLM-R). MMBERT outperforms the smaller

variants: 73.6 XNLI vs Multilingual MiniLM’s 71.2, even coming close to XLM-R’s 74.6 despite having $\sim 1/3$ of the amount of non-embed parameters.

Retrieval Results We train all models on MS MARCO (English, §B) and evaluate on both English and multilingual benchmarks from MMTEB v2. We also evaluate on code using the CoIR benchmark.

We again see large gains in English MTEB v2 (Table 7) with even MMBERT small outperforming mGTE and XLM-R. MMBERT outperforms mGTE (the next closest) with an average of 53.9 vs 52.7, and even performs similarly to ModernBERT’s 53.8 average.

For multilingual MTEB v2 (Table 8), we find that both MMBERT models score about 1.5 points better on average than their similarly sized counterparts (i.e. 54.1 avg for MMBERT base vs XLM-R’s 52.4).

On code retrieval tasks (CoIR, Table 6) we see that MMBERT performs significantly better than any other massively multilingual model (42.2 MMBERT base average vs XLM-R’s 33.6), but underperforms compared to EuroBERT-210m (45.3 average). This is likely due to EuroBERT’s use of the higher quality Stack v2, which is not openly available.

Overall, we see that MMBERT is an improved drop-in replacement for XLM-R, and that even MMBERT small can come close to XLM-R’s performance.

4.2. Comparison to EuroBERT

As EuroBERT was only trained on 15 languages, it has a disadvantage on these massively multilingual evaluations and was excluded due to its low performance. Instead, we directly compare EuroBERT only on languages it was trained on. We show scores on these selected languages for

⁷We do not compare with RTD models. These models have consistently been shown to be significantly worse for embeddings. We validate this for mDeBERTa (He et al., 2023) in App C, where it is more than 11 pts worse.

Table 7. MTEB v2 English results. Average is done over categories. Best model in the section is bolded. See Appendix B for sweep details. We see that both MMBERT models significantly outperform all multilingual models and MMBERT base even ties ModernBERT.

	Model Name	Pair Class.	Class.	STS	Retrieve	Cluster	Rerank	Summ.	Avg
Small	Multilingual MiniLM	77.3	59.3	71.9	35.0	36.9	42.4	19.3	48.9
	mDistilBERT	75.9	59.0	72.4	38.4	38.3	42.7	19.4	49.4
	MMBERT small	78.9	62.1	74.1	40.7	40.8	44.1	23.6	52.1
Base	EuroBERT 210m	79.5	62.3	73.6	43.0	40.3	43.3	21.5	51.9
	XLM-R base	79.3	63.9	73.4	39.8	39.3	44.2	24.2	52.0
	mGTE base	79.9	64.3	74.4	43.7	40.2	43.7	23.0	52.7
	ModernBERT base	80.5	65.6	75.5	44.8	43.3	44.1	22.6	53.8
	MMBERT base	80.2	64.8	74.8	44.9	41.7	44.9	26.0	53.9

Table 8. MTEB v2 multilingual results. Average is done over categories. See Appendix B for sweep details. We see that both MMBERT models significantly outperform all other models for their size.

	Model Name	Bitext Mining	Pair Class.	Class.	STS	Retrieve	Multilabel Class.	Cluster	Rerank	Avg
Small	mDistilBERT	37.2	76.6	50.3	62.1	36.7	14.6	37.5	62.2	47.1
	Multilingual MiniLM	46.6	76.7	51.0	64.6	35.6	14.0	34.9	64.1	48.4
	MMBERT small	50.5	77.4	50.4	64.8	41.9	15.5	38.7	66.1	50.7
Base	XLM-R base	56.6	78.2	54.5	66.0	43.0	15.5	38.1	67.6	52.4
	mGTE base	52.6	78.5	53.1	66.2	46.4	17.2	38.8	69.1	52.7
	MMBERT base	59.2	79.2	53.6	67.1	45.8	17.5	40.2	69.9	54.1

Table 9. Comparing EuroBERT and MMBERT on **EuroBERT’s in-distribution languages** (excluding others) on XNLI and PAWS-X. We see the MMBERT improves even on these languages. Dashes indicate a language not tested by the benchmark.

XNLI												
Model	ar	de	en	es	fr	hi	ja	ru	tr	vi	zh	Avg
EuroBERT 210m	66.8	72.5	84.7	76.7	75.9	60.6	–	72.6	66.4	70.1	72.7	71.9
MMBERT small	71.4	77.0	85.8	79.9	79.0	67.9	–	75.6	72.6	75.3	73.2	75.8
MMBERT base	74.5	79.6	85.9	81.2	80.2	71.1	–	77.0	73.8	76.2	77.7	77.7
PAWS-X												
EuroBERT 210m	–	87.8	94.9	89.6	90.0	–	77.6	–	–	–	80.2	86.7
MMBERT small	–	89.8	95.3	90.5	90.6	–	80.7	–	–	–	82.9	88.3
MMBERT base	–	90.7	95.8	90.2	92.0	–	81.8	–	–	–	83.6	89.0

XNLI and PAWS-X.⁸ Table 9 shows that MMBERT base and small outperform EuroBERT-210m on these languages as well (i.e. 74.5 F1 on Arabic XNLI for MMBERT base vs 66.8 F1 for EuroBERT, etc.).

4.3. Comparison to Similar-Sized SoTA Decoders

Although numerous previous works (Weller et al., 2025; Gisserot-Boukhlef et al., 2025) have shown that encoder-only models significantly outperform decoder models within the same size range (or even one order-of-magnitude big-

⁸Since the official EuroBERT model does not have a “ForQuestionAnswering” or “ForMultipleChoice” class available we select two tasks that work for classification.

ger), we also test this by running the recent Gemma 3 270M (Team et al., 2025) model on the same classification tasks. Using the same hyperparameter sweep detailed previously, it scores 69.0 on XNLI and averages 82.9 on GLUE. Notably, this is much worse than even MMBERT small, again showing the benefits of encoder-only models for these tasks.⁹

⁹We note that an embedding version of Gemma 3 270M, *EmbeddingGemma*, outperforms other models on MTEB, but crucially they fine-tune the model using a proprietary 320 billion token training set. They would likely see increased performance over EmbeddingGemma if they simply fine-tuned MMBERT with the same data. However, as their data is not publicly available we cannot run this comparison.

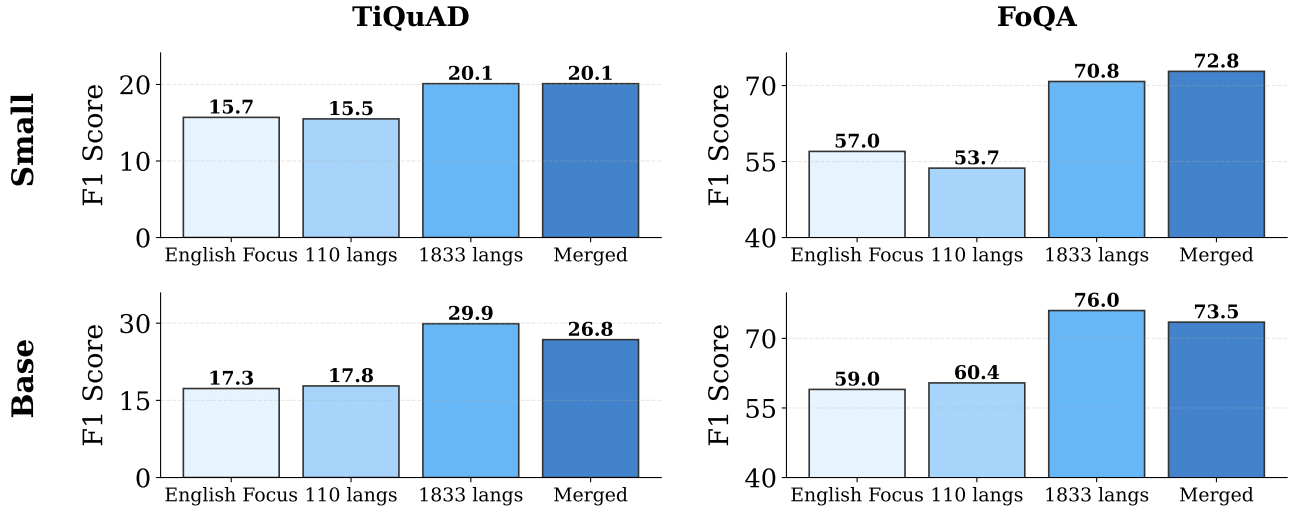


Figure 2. Performance of models using different decay phases on two languages (Tigrinya and Faroese) only added during the decay phase. We see that MMBERT with the 1833 language decay phase shows rapid performance improvements despite only having the models in the last 100B tokens of training. The final MMBERT models shows improvements by merging together checkpoints.

4.4. Annealed Language Learning

We test whether our decision to include more languages at the decay phase significantly improves model performance. We do this by selecting evaluation datasets that test the model on a language learned exclusively during the decay phase. However, since these languages are very low-resource there are not many high-quality evaluation datasets available. Thus, we test only two languages that have high quality evaluation data and are commonly¹⁰ used: TiQuAD for Tigrinya (Gaim et al., 2023) and FoQA for Faroese (Simonsen et al., 2025). Note that these languages are related to higher resource languages present in other training stages, which may result in positive transfer. We evaluate in the same way as XQuAD (e.g. zero-shot from English SQuAD) and compare scores for the different decay mixtures and the final merged model.

Figure 2 shows that there is a significant jump in performance for the variants that included the language: a 68% increase for base (12.1 absolute F1) on Tigrinya (110 to 1833 langs) and a 26% increase (15.4 absolute F1) for Faroese. On FoQA, which benchmarked larger LMs, it even outperforms Google’s Gemini 2.5 Pro by 6 pts (69.8) and OpenAI’s o3 by 8.3 points (67.7). Even MMBERT small outperforms these giant LMs, showing the benefits of models focused on multilingual mixtures.

Thus we find that annealed language learning significantly boosts low-resource language performance. We also see that the model merging allowed the model to retain most of its performance despite merging with English and higher-resource focused versions.

¹⁰Are in other benchmarks, i.e. ScandEval (Nielsen, 2023).

4.5. Efficiency

MMBERT models are significantly faster than any previous multilingual encoder-only model. Figure 3 benchmarks this using the best settings available for each model using the `transformers` (Wolf et al., 2020) implementation, e.g. by installing `xformers` or `flash-attention` when helpful. We find that MMBERT base is more than 2x faster on variable sequences and significantly faster on long context lengths ($\sim 4x$). For MMBERT small results are roughly 2x faster than MMBERT base and again roughly 2x faster than similarly-sized other multilingual models. We note that previous multilingual models such as Multilingual MiniLM and XLM-R cannot go past 512 tokens whereas MMBERT can perform at up to 8192 tokens – and can do so as fast as other models are at 512 tokens.

5. Conclusion

We introduce MMBERT, a modern multilingual encoder trained on 3T tokens and 1833 languages. We introduce several novel elements in training: an inverse masking schedule and a cascading annealed language learning schedule for multilingual data that progressively lower the language sampling temperature while also increasing the number of languages. MMBERT improves over the previous generation of multilingual encoders such as XLM-R while being a drop-in replacement. We further show that leaving out low-resource languages until the decay phase allows us to rapidly learn the languages on minute amounts of data, improving performance to levels past even SoTA large decoders. We open-source our models, data, and checkpoints to encourage

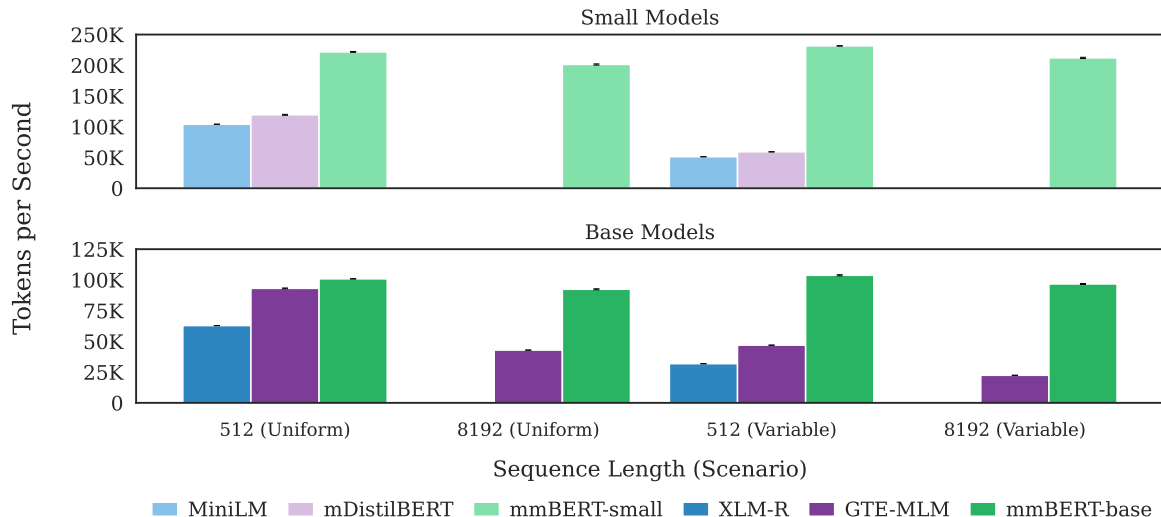


Figure 3. Throughput efficiency for various sequence lengths and variable input length (top row is small models, bottom row is base models). MMBERT is faster than previous models even when using 8x as many tokens, often 2-4x as fast. MMBERT models are more efficient because of the use of Flash Attention 2 and unpadding techniques it inherits from the ModernBERT configuration. Empty bars indicate that the model cannot use that sequence length (e.g. over 512 tokens). The error bars show standard error over five random seeds.

future research and enable continued pre-training for new data and domains.

Impact Statement

The goal of this work is to advance multilingual encoder models. We do not expect the ethical impact of this model to be substantially different from similar model releases. Nevertheless, we highlight a few specific limitations of this work. Although this work incorporates more multilingual resources than earlier work (and shows improvements on lower resourced languages), it still depends on the upstream datasets for language identification quality and coverage. The upstream resources have been filtered but may still contain noisy, biased, or harmful content that was not discovered. We believe these risks are further mitigated in our models as they are encoders that cannot be straightforwardly used for generation (c.f. a model trained with decoder objectives). We also note that tokenization choices made early in this project may have introduced weaknesses for tasks like NER and POS.

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A. Architecture Details

Common architecture details are in Table 10. One other area of difference between the two models is that MMBERT small used a learning rate warmup of 4B tokens and a batch size warmup of 100B tokens. MMBERT base had a learning rate warmup of 3B tokens and a batch size warmup of 60B tokens.

Parameter	Value
Learning Rate	8e-4
Weight Decay	8e-5
Training Batch Size	4.7M tokens
Vocabulary Size	256,000
Max Sequence Length	1024->8192
Tokenizer	Gemma 2
Attention Layer	RoPE
Attention Dropout	0.0
Attention Output Bias	false
Attention Output Dropout	0.1
Attention QKV Bias	false
Transformer Layer	prenorm
Embedding Dropout	0.0
Embedding Norm	true
Final Norm	true
Skip First PreNorm	true
MLP Dropout	0.0
MLP Input Bias	false
MLP Layer Type	GLU
MLP Output Bias	false
Normalization	LayerNorm
Norm Epsilon	1e-12
Norm Bias	false
Hidden Activation	GELU
Head Pred Activation	GELU
Activation Function	GELU
Padding	unpadded
Rotary Embedding Base	10k -> 160k
Rotary Embedding Interleaved	false
Allow Embedding Resizing	true
Sliding Window	128
Global Attention Every N Layers	3
Unpad Embeddings	true

Table 10. Common Configuration Parameters. Note that MMBERT small had to lower the LR/WD to half the initial value after 1.2T tokens, due to it plateauing early.

B. Hyperparameter and Compute Details

Compute We use a mix of H100 and L40s for training and inference, using the L40s mainly for the smaller model inference. We use 8xH100s for roughly 10 days to train the small version and 8xH100s for 40 days for the base version. For evaluation, each experiment takes roughly 1-2 hours to run for a given setting.

NLU Sweep We sweep over seven LR_s ({2e-5, 3e-5, 4e-5, 5e-5, 6e-5, 7e-5, 8e-5}) and four epoch options ({1, 2, 3, 5, 10}) using a batch size of 32 and a warmup ratio of 0.06 (following mGTE). We select the best result for each model on

each task in an oracle fashion. Most models had the best score in the 2e-5 or 3e-5 range with varying amounts of epochs according to the task.

Embedding Sweep We sweep four LR, following Weller et al. (2025) {1e-4, 3e-4, 5e-4, 7e-4} and use the best one. We train with SentenceTransformers (Reimers & Gurevych, 2019) using 1.25M hard triplets for one epoch on MS MARCO data (Bajaj et al., 2016) from sentence-transformers/msmarco-co-condenser-margin-mse-sym-mnrl-mean-v1. We then evaluate using MTEB (Enevoldsen et al., 2025) and select the LR that performed the best. In all cases the best was using the 1e-4 LR (except for MiniLM which was 5e-4).

FoQA and TiQuAD We use a partial sweep from the NLU section, using just the two LR that performed best from the original sweep (2e-5, 3e-5) while also varying epochs.

C. Comparison with DeBERTa

Both Ettin and ModernBERT found that comparable RTD-trained models show significantly worse performance on embedding tasks. We thus exclude them from our main analysis.

We validate this by running mDeBERTa on MTEB tasks. On the multilingual benchmark, we find it has an average of 42.5, more than 11 points worse than mMBERT’s 54.1. mDeBERTa scores 48.6 on the English MTEB v2, again much worse than any comparable model (including our mMBERT small). Thus, RTD-trained models may do well at classification but they do so at the expense of embedding tasks which are a main use case of encoder-only models.

D. Language Distributions

The data per language is found in Table 11. We use an inverse temperature sampling with values 0.7, 0.5, and 0.3 respectively. As only 80 rows would fit on the page, the full data is available on the Github in CSV form.

E. LLM Usage

LLMs were not used for paper writing, but were used for coding assistance and title brainstorming. All code was human verified.

Lang	Pretrain		Mid-Training		Decay-Eng		Decay-Cont		Decay-All		Total
	Value	%	Value	%	Value	%	Value	%	Value	%	
eng	686.4	34.51%	73.0	12.14%	18.7	17.89%	13.6	13.19%	11.6	10.15%	803.3
code	102.0	5.13%	20.0	3.33%	3.3	3.12%	3.3	3.15%	6.3	5.51%	134.8
rus	115.4	5.80%	30.2	5.02%	2.8	2.72%	3.1	2.98%	3.0	2.61%	154.5
deu	87.3	4.39%	24.7	4.11%	2.5	2.44%	2.8	2.75%	2.8	2.46%	120.2
spa	89.8	4.51%	25.2	4.19%	2.5	2.38%	2.7	2.62%	2.7	2.37%	122.8
fra	79.8	4.01%	23.1	3.85%	2.4	2.28%	2.6	2.50%	2.6	2.23%	110.4
cmn	103.7	5.21%	27.9	4.65%	2.5	2.38%	2.6	2.47%	2.5	2.21%	139.2
ita	57.3	2.88%	18.3	3.05%	2.0	1.96%	2.2	2.12%	2.1	1.88%	82.0
por	46.9	2.36%	16.5	2.75%	1.9	1.83%	2.0	1.98%	2.0	1.76%	69.4
jpn	58.6	2.95%	18.6	3.10%	2.0	1.87%	2.0	1.96%	2.0	1.75%	83.2
nld	37.2	1.87%	13.4	2.23%	1.8	1.68%	2.0	1.90%	1.9	1.70%	56.2
pol	36.5	1.84%	13.2	2.20%	1.7	1.65%	1.9	1.86%	1.9	1.68%	55.3
swe	18.6	0.94%	8.1	1.36%	1.4	1.37%	1.7	1.62%	1.6	1.42%	31.5
ind	32.5	1.63%	12.2	2.03%	1.5	1.47%	1.6	1.56%	1.6	1.41%	49.4
kor	28.4	1.43%	11.1	1.84%	1.5	1.39%	1.5	1.47%	1.5	1.31%	43.9
arb	21.2	1.07%	9.0	1.50%	1.4	1.30%	1.5	1.43%	1.7	1.48%	34.7
ces	22.7	1.14%	9.4	1.57%	1.4	1.32%	1.5	1.41%	1.4	1.24%	36.4
tur	24.9	1.25%	10.1	1.68%	1.4	1.32%	1.4	1.40%	1.4	1.25%	39.3
ukr	17.5	0.88%	7.8	1.30%	1.3	1.22%	1.4	1.39%	1.4	1.25%	29.4
fas	23.3	1.17%	9.6	1.60%	1.4	1.30%	1.4	1.39%	1.4	1.24%	37.1
vie	22.2	1.12%	9.3	1.54%	1.3	1.27%	1.4	1.37%	1.4	1.23%	35.6
ron	22.4	1.12%	9.3	1.55%	1.3	1.26%	1.4	1.32%	1.4	1.25%	35.8
nob	20.9	1.05%	8.9	1.48%	1.3	1.22%	1.3	1.30%	1.3	1.17%	33.8
hun	20.2	1.02%	8.7	1.45%	1.3	1.20%	1.3	1.27%	1.3	1.14%	32.8
fin	14.9	0.75%	7.0	1.16%	1.2	1.12%	1.3	1.24%	1.3	1.10%	25.5
dan	19.2	0.96%	8.4	1.40%	1.2	1.15%	1.2	1.20%	1.2	1.08%	31.2
tha	18.2	0.92%	8.1	1.34%	1.2	1.14%	1.2	1.19%	1.2	1.06%	29.9
ell	16.4	0.83%	7.5	1.25%	1.2	1.12%	1.2	1.19%	1.2	1.05%	27.5
bul	12.9	0.65%	6.3	1.05%	1.0	0.99%	1.1	1.04%	1.1	0.93%	22.4
slk	12.2	0.61%	6.1	1.01%	1.0	0.96%	1.0	0.99%	1.0	0.89%	21.3
hrv	11.3	0.57%	5.8	0.96%	1.0	0.92%	1.0	0.96%	1.0	0.87%	20.0
srp	0.0	0.00%	3.8	0.63%	1.0	0.92%	1.0	0.93%	0.9	0.82%	6.7
heb	8.3	0.42%	4.6	0.77%	0.9	0.85%	0.9	0.92%	0.9	0.81%	15.7
hin	10.0	0.50%	5.3	0.88%	0.9	0.87%	0.9	0.90%	1.2	1.04%	18.3
lit	8.8	0.44%	4.8	0.80%	0.9	0.83%	0.9	0.86%	0.9	0.78%	16.2
ekk	7.0	0.35%	4.1	0.68%	0.8	0.79%	0.9	0.84%	0.8	0.74%	13.7
slv	7.9	0.40%	4.5	0.74%	0.8	0.80%	0.9	0.83%	0.8	0.74%	14.9
uzn	0.0	0.00%	2.4	0.40%	0.8	0.80%	0.8	0.81%	0.7	0.63%	4.8
bos	0.0	0.00%	4.7	0.78%	0.8	0.79%	0.8	0.80%	0.8	0.72%	7.2
ben	6.5	0.33%	3.9	0.64%	0.8	0.74%	0.8	0.79%	0.9	0.78%	12.8
cat	0.0	0.00%	4.4	0.74%	0.8	0.77%	0.8	0.77%	0.8	0.70%	6.8
zsm	6.3	0.32%	3.8	0.63%	0.7	0.72%	0.8	0.75%	0.8	0.73%	12.4
lvs	6.2	0.31%	3.7	0.62%	0.8	0.72%	0.8	0.75%	0.8	0.66%	12.2
azj	4.8	0.24%	3.1	0.52%	0.7	0.66%	0.7	0.69%	0.7	0.65%	10.0
als	0.0	0.00%	2.9	0.49%	0.7	0.65%	0.7	0.65%	0.6	0.54%	4.9
kaz	3.0	0.15%	2.2	0.37%	0.6	0.56%	0.6	0.61%	0.6	0.54%	7.0
tam	3.1	0.15%	2.3	0.38%	0.6	0.56%	0.6	0.60%	0.7	0.61%	7.2
urd	3.6	0.18%	2.6	0.43%	0.6	0.58%	0.6	0.60%	0.9	0.74%	8.3
kat	2.5	0.13%	1.9	0.32%	0.5	0.51%	0.6	0.56%	0.6	0.50%	6.1
isl	0.0	0.00%	2.1	0.35%	0.6	0.53%	0.6	0.53%	0.5	0.44%	3.7
mkd	2.6	0.13%	2.0	0.33%	0.5	0.50%	0.5	0.53%	0.5	0.47%	6.2
afr	2.7	0.13%	2.1	0.34%	0.5	0.50%	0.5	0.52%	0.5	0.46%	6.3
tel	2.1	0.10%	1.7	0.29%	0.5	0.48%	0.5	0.52%	0.6	0.49%	5.4
mya	1.8	0.09%	1.6	0.26%	0.5	0.48%	0.5	0.51%	0.5	0.42%	4.9
ary	0.0	0.00%	2.1	0.35%	0.5	0.51%	0.5	0.51%	0.5	0.45%	3.7
bel	0.0	0.00%	1.8	0.29%	0.5	0.50%	0.5	0.51%	0.4	0.39%	3.3
mar	2.6	0.13%	2.0	0.34%	0.5	0.48%	0.5	0.50%	0.5	0.48%	6.2
fil	2.6	0.13%	2.0	0.34%	0.5	0.48%	0.5	0.50%	0.5	0.44%	6.1
glg	2.2	0.11%	1.8	0.30%	0.5	0.47%	0.5	0.49%	0.5	0.44%	5.5
mal	2.0	0.10%	1.7	0.28%	0.5	0.45%	0.5	0.49%	0.6	0.49%	5.2
npi	2.5	0.13%	2.0	0.33%	0.5	0.47%	0.5	0.49%	0.5	0.44%	6.0
lat	0.0	0.00%	1.4	0.24%	0.5	0.45%	0.5	0.45%	0.4	0.34%	2.7
bod	0.0	0.00%	0.8	0.13%	0.5	0.45%	0.5	0.45%	0.2	0.21%	1.9
khk	1.7	0.08%	1.5	0.24%	0.4	0.43%	0.5	0.44%	0.4	0.37%	4.4
pan	0.0	0.00%	1.2	0.20%	0.4	0.43%	0.4	0.44%	0.4	0.33%	2.5
gmh	0.0	0.00%	1.6	0.26%	0.4	0.43%	0.4	0.43%	0.4	0.37%	2.9
guj	1.7	0.09%	1.5	0.26%	0.4	0.41%	0.4	0.43%	0.5	0.39%	4.6
anp	0.0	0.00%	1.6	0.27%	0.4	0.42%	0.4	0.42%	0.4	0.38%	2.9
hye	0.0	0.00%	1.2	0.19%	0.4	0.41%	0.4	0.41%	0.3	0.30%	2.4
rmy	0.0	0.00%	1.0	0.17%	0.4	0.40%	0.4	0.41%	0.3	0.28%	2.2
eus	0.0	0.00%	1.4	0.23%	0.4	0.40%	0.4	0.40%	0.4	0.34%	2.6
kan	0.0	0.00%	1.4	0.23%	0.4	0.39%	0.4	0.39%	0.4	0.35%	2.6
cym	0.0	0.00%	1.1	0.18%	0.4	0.39%	0.4	0.39%	0.3	0.30%	2.2
khm	1.4	0.07%	1.3	0.21%	0.4	0.38%	0.4	0.39%	0.4	0.34%	3.9
swl	1.3	0.06%	1.2	0.20%	0.4	0.37%	0.4	0.38%	0.4	0.33%	3.7
sin	1.2	0.06%	1.1	0.19%	0.4	0.36%	0.4	0.38%	0.8	0.66%	3.9
ars	0.0	0.00%	1.2	0.20%	0.4	0.37%	0.4	0.37%	0.4	0.32%	2.3
nno	0.0	0.00%	1.1	0.19%	0.4	0.35%	0.4	0.35%	0.3	0.30%	2.2
bew	0.0	0.00%	1.1	0.19%	0.4	0.35%	0.4	0.35%	0.3	0.30%	2.2
ory	0.0	0.00%	0.9	0.15%	0.4	0.34%	0.4	0.35%	0.3	0.26%	1.9
Total	1989.0	100.00%	600.8	100.00%	104.4	100.00%	103.3	100.00%	114.2	100.00%	2911.8

Table 11. Language data (in billions, rounded to nearest 100M) with stage percentages for the first 80 languages included. More languages would not fit on the page, see the Github for the full CSV details.